

Ring

An algebraic structure $(R, +, \times)$, where R is a non-empty set with two binary operation $+$ & $\times$ defined on R is called ring if it holds the following properties:

1. Closure Under Addition & Multiplication

$$(a + b) \in R \quad [a, b \in R] \quad \& \quad (a \times b) \in R$$

2. Associativity of Under Addition & Multiplication

$$a, b, c \in R$$

$$(a + b) + c = a + (b + c) \quad \& \quad (a \times b) \times c = a \times (b \times c)$$

3. Additive Identity & Multiplicative Identity
(Not Strictly Required)

$$a + 0 = a, \quad a \in R \quad \& \quad a \times 1 = a$$

(Not required for basic ring)

4. Additive Inverse & Multiplicative Inverse
(Not Strictly Required)

$$a + (-a) = 0 = e \quad \& \quad a \times a^{-1} = 1 = e$$

(Not required for basic ring)

5. Commutative Under Addition & Multiplication
(Not Strictly Required)

$$a + b = b + a \quad \& \quad a \cdot b = b \cdot a$$

(Not required for basic ring)

6. Distributive Law

Multiplication must distribute over addition in both ways:

Left Distributed Law - $a \times (b + c) = (a \times b) + (a \times c)$

Right Distributed Law - $(a + b) \times c = (a \times c) + (b \times c)$

Q. If $R = \{1, 2, 3, 4, 0\}$ ($R, +_5, \times_5$) define, check the given structure is ring or not?

Solⁿ:

$R = \{0, 1, 2, 3, 4\}$ w.r.to $+_5$ & $\times_5$ (or $\cdot_5$)

For $+_5$

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Closure, $(a +_5 b) \in R$ i.e. 0, 1, 2, 3 & 4 belongs to R .

Associativity, $(a +_5 b) +_5 c = a +_5 (b +_5 c)$
 $\forall a, b, c \in R$

$$(0 +_5 1) +_5 2 = 0 +_5 (1 +_5 2) \\ \Rightarrow 3 = 3 \text{ (True)}$$

Identity, $a +_5 0 = a \quad \forall a \in R$ [True]

Inverse, $a +_5 (-a) = 0$ & $(-a) \in R$

For 0, $-0 = 0 \in R$ & $0 +_5 0 = 0$

For 1, $-1 = 4 \in R$ & $1 +_5 4 = 0$

For 2, $-2 = 3 \in R$ & $2 +_5 3 = 0$

For 3, $-3 = 2 \in R$ & $3 +_5 2 = 0$

For 4, $-4 = 1 \in R$ & $4 +_5 1 = 0$

Commutative, $a +_5 b = b +_5 a \quad \forall a, b \in R$
[True]

For $\times_5$

Closure, $(a \times_5 b) \in R$ i.e. $0, 1, 2, 3$ & $4 \in R$.

Associativity, $a \times_5 (b \times_5 c) = (a \times_5 b) \times_5 c$
 $\forall a, b, c \in R$

For Distributive Laws

Left Distributive Law, $a \times_5 (b +_5 c) = (a \times_5 b) +_5 (a \times_5 c)$

Right Distributive Law, $(a +_5 b) \times_5 c = (a \times_5 c) +_5 (b \times_5 c)$

Since, all ring conditions are satisfied, the given structure is a ring.

Types of Ring

1. Commutative Ring: A ring where multiplication is commutative.
i.e. $a \cdot b = b \cdot a \quad \forall a, b \in R$
2. Ring with Unity (Unital Ring): A ring that contains a multiplicative identity element (denoted as 1) such that $a \cdot 1 = 1 \cdot a = a \quad \forall a \in R$.
3. Ring with Zero Divisors: A ring containing non-zero elements ~~and~~ a & b such that their product is zero.
i.e. $0 \neq R$ & $a \cdot b = 0 \quad \forall a, b \in R$.
4. Ring without Zero Divisors: A ring where product of two non-zero elements a & b is never zero and product is zero only when at least one of the elements is zero.

$$\text{i.e. } a \cdot b \neq 0, \quad a \neq 0 \text{ \& } b \neq 0$$

$$a \cdot b = 0, \quad a = 0 \text{ or } b = 0 \quad \forall a, b \in R$$

Q. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ & $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, then are two non-zero element then determine ring is zero divisor or not?

Solⁿ:

$$A \cdot B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Thus, ring is zero divisor ring.

Field

A field $(F, +, \times)$ [or $(F, +, \cdot)$] is a commutative ring with unity in which every non-zero element has a multiplicative inverse. The conditions for field are:

Multiplicative Identity: There exists an element $1 \in F$ such that $a \cdot 1 = a \quad \forall a \in F$.

Multiplicative Commutativity:

$$a \cdot b = b \cdot a \quad \forall a, b \in F.$$

Multiplicative Inverse: For every $a \in F$ where $a \neq 0$, there exists an element $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$.

Ring vs Field

Feature	Ring $(R, +, \cdot)$	Field $(F, +, \cdot)$
Additive Structure	Abelian group (closure, associative, identity 0 , inverses $-a$, commutative).	Abelian group (identical to a ring).
Multiplicative Associativity	Always required.	Always required.
Distributive Laws	Multiplication distributes over addition.	Multiplication distributes over addition.
Multiplicative Commutativity	Not required.	Strictly required. ($a \cdot b = b \cdot a$)
Multiplicative Identity	Optional.	Strictly required.
Multiplicative Inverse	Generally absent.	Every non-zero element must have an inverse.
Zero Divisors	Can exist. ($a \cdot b = 0, a, b \neq 0$)	Never exist. ($a \cdot b = 0, a = 0$ or $b = 0$)